ALETI VENKATESH

Backend Engineer | Python | FastAPI | PostgreSQL | GenAI

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PROFESSIONAL SUMMARY

Backend engineer with 3 years at Accenture, building Python and FastAPI services, REST APIs and PostgreSQL-backed systems for enterprise clients. Experienced in GenAI and LLM applications, including RAG pipelines, multi-agent workflows and Azure OpenAI integrations, along with database design. Also worked on API security fixes, telemetry and distributed tracing, and deployments through Azure Pipelines.

CORE SKILLS

Backend: Python, FastAPI, REST APIs, SQL, PostgreSQL, SQL Server

GenAI & LLMs: LLM applications, RAG, AI Agents, Google ADK, LangChain, LangGraph, CrewAI, Prompt Engineering

Cloud & Azure AI: Azure OpenAI, Azure AI Search, Azure Blob Storage, Azure Document Intelligence, GCP Vertex AI

Engineering: Git, Docker, Postman, Azure Pipelines, SSE / Streaming, Telemetry & Tracing

Security: JWT, CORS, Authorization, IDOR Remediation

PROFESSIONAL EXPERIENCE

Accenture — Packaged App Development Analyst

Oct 2023 – Present

Hyderabad, India

- Built backend services and APIs in Python and FastAPI across four enterprise projects, moving from functional API work into GenAI and agentic systems.
- Developed document processing pipelines, RAG workflows and agentic workflow orchestration on Azure AI and Google Cloud.
- Worked on API security, telemetry and deployments through Azure Pipelines.

KEY PROJECTS

CGS: Audit and ReconAgent AI Platforms

Apr 2026 – Present

Stack: Python, FastAPI, Google ADK, Gemini (Vertex AI), PostgreSQL

- Built the Python FastAPI agentic service for an audit automation platform, orchestrating multiple Google ADK agents on Gemini for evidence collection, PBC reporting and auditor query resolution.
- Enforced structured LLM output with Pydantic schemas, with retries and a graceful fallback when the model was unavailable.
- Designed the PostgreSQL schema and end-to-end workflow for audit samples, evidence, PBC reports and query tracking.
- Built ReconAgent features including scenario-based processing, BlackLine reporting and monthly amortisation integration.
- Cut the audit evidence cycle from about a week to under a day.

AIR 2.0: Agentic Workflow Orchestration Platform

Sep 2025 – Mar 2026

Stack: Python, FastAPI, LangGraph, SSE

- Built FastAPI endpoints for workflows, workflow instances, an agent registry and an API registry to support agentic workflow orchestration.
- Implemented API node execution with retry logic and error handling, integrating workflow runs with agent APIs.
- Improved workflow telemetry and distributed tracing with live node status and error categorisation for monitoring.
- Built SSE streaming for real-time visibility into workflow execution.
- Fixed API security issues, including CORS restrictions, authorization and IDOR vulnerabilities, and JWT verification.

CPF: Azure AI Document Platform *Feb 2025 – Aug 2025*

Stack: Python, FastAPI, SQL Server, Azure OpenAI, Azure AI Search, Document Intelligence, CrewAI

- Built a CrewAI agentic workflow where an orchestrator agent coordinates specialist agents to generate quarterly business reports automatically.
- Built Python FastAPI backend services on SQL Server for CRUD operations and business REST endpoints.
- Developed a document processing pipeline on Azure Blob Storage, Document Intelligence and AI Search, handling 30 to 40 documents per batch through chunking, metadata generation and vector ingestion.
- Implemented RAG workflows with Azure OpenAI for summarisation, including prompt tuning and feedback summarisation.
- Delivered PPTX and PDF export, plus automated email delivery of generated reports.

Gen ISP: Enterprise Backend APIs *Feb 2024 – Jan 2025*

Stack: SQL Server, REST APIs

- Developed backend REST APIs supporting core business workflows for an enterprise application.
- Implemented API logic aligned to business requirements and validated it through API and application testing.

CERTIFICATIONS

- Claude Certified Architect — Foundations (CCA-F), Anthropic — May 2026
- GitHub Copilot (GH-300), GitHub — Sep 2025
- Microsoft Certified: Azure Fundamentals (AZ-900), Microsoft — Apr 2024
- Microsoft Certified: Power Platform Fundamentals (PL-900), Microsoft — Feb 2024

EDUCATION

B.Tech, Computer Science and Engineering *2019 – 2023*

Raghu Engineering College, JNTU Kakinada | CGPA: 9.43

ACADEMIC PROJECT & PUBLICATION

Course Recommendation System using Machine Learning (Python) — Built a recommender suggesting courses based on academic and career objectives, including course prerequisites and resources. Publication: journal-dogorangsang.in/no_1_Online_23/13_apr.pdf